

223283 Seed Weight Grid

These configs scan the same seed-weight mixtures used for the 1111708 grid, but for organism 223283.

Weight meaning:

- `alpha` : InterPro seed weight
- `beta` : HMMER seed weight
- `gamma` : Foldseek seed weight
- `delta` : PLM seed residual, computed as `1 - alpha - beta - gamma`

Use `223283.conf` as the current baseline. The grid configs assume that this baseline has already produced the 223283 graph collection, homology graph, InterPro seed, HMMER seed, Foldseek assignments, and PLM assignments. Each grid config reuses those baseline artifacts and keeps `combined_graph = compute` so S2F learns graph-combination coefficients from each weighted seed target.

Config	alpha	beta	gamma	delta	Purpose
<code>223283.conf</code>	0.60	0.10	0.20	0.10	Current baseline
<code>223283_w_equal.conf</code>	0.25	0.25	0.25	0.25	Equal four-seed mixture
<code>223283_w_interpro_heavy.conf</code>	0.70	0.10	0.10	0.10	InterPro-heavy
<code>223283_w_hmmer_heavy.conf</code>	0.20	0.50	0.20	0.10	HMMER-heavy
<code>223283_w_foldseek_heavy.conf</code>	0.20	0.10	0.60	0.10	Foldseek-heavy
<code>223283_w_plm_heavy.conf</code>	0.20	0.10	0.10	0.60	PLM-heavy
<code>223283_w_no_interpro.conf</code>	0.00	0.33	0.33	0.34	Leave out InterPro
<code>223283_w_no_hmmer.conf</code>	0.33	0.00	0.33	0.34	Leave out HMMER
<code>223283_w_no_foldseek.conf</code>	0.33	0.33	0.00	0.34	Leave out Foldseek
<code>223283_w_no_plm.conf</code>	0.34	0.33	0.33	0.00	Leave out PLM
<code>223283_w_foldseek_plm.conf</code>	0.00	0.00	0.50	0.50	Foldseek plus PLM only
<code>223283_w_interpro_hmmer.conf</code>	0.90	0.10	0.00	0.00	Original-style InterPro plus HMMER

Config	alpha	beta	gamma	delta	Purpose
223283_w_interpro_foldseek.conf	0.50	0.00	0.50	0.00	InterPro plus Foldseek only
223283_w_interpro_plm.conf	0.50	0.00	0.00	0.50	InterPro plus PLM only
223283_w_hmmer_foldseek.conf	0.00	0.50	0.50	0.00	HMMER plus Foldseek only
223283_w_hmmer_plm.conf	0.00	0.50	0.00	0.50	HMMER plus PLM only
223283_w_interpro_only.conf	1.00	0.00	0.00	0.00	InterPro-only baseline
223283_w_hmmer_only.conf	0.00	1.00	0.00	0.00	HMMER-only baseline
223283_w_foldseek_only.conf	0.00	0.00	1.00	0.00	Foldseek-only baseline
223283_w_plm_only.conf	0.00	0.00	0.00	1.00	PLM-only baseline

Run the baseline first with:

```
python S2F.py predict --run-config conf/weight_grid/done/223283/223283.conf
```

Run one grid config with:

```
python S2F.py predict --run-config conf/weight_grid/done/223283/223283_w_equal.conf
```